

AI Sales Predictor

Model Deployment: SuperKart PGP-AIML-BA-UTA-Mar25-E

Date 11/02/2025

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Executive Summary

Business Need

Sales forecasting is a vital function that supports strategic and operational decisions. It enables accurate **demand planning**, effective **budgeting and financial management**, and provides a basis for **performance evaluation** by comparing actual versus projected sales. Forecasting also helps anticipate **market fluctuations**, ensuring SuperKart can proactively manage resources, inventory, and potential risks.

Company Context

SuperKart is a leading retail chain operating multiple supermarkets and food marts across various cities. The company offers a diverse product range, serving different regional markets. To improve **inventory optimization** and **regional sales strategies**, SuperKart aims to forecast its **next quarter's sales revenue** with higher accuracy.

Project Overview

To achieve this goal, SuperKart partnered with a **data science firm** to develop a robust predictive model using its **historical sales data**. The objective was to build a model that could be easily integrated into decision-making processes and made accessible across all store locations.

Model Development

The data science team experimented with multiple machine learning algorithms and identified **XGBoost** as the most suitable model due to its superior accuracy and reliability. This model effectively captures complex relationships between features influencing sales.

Executive Summary

Deployment and Accessibility

To ensure seamless accessibility and scalability, the forecasting model was **containerized using Docker** and deployed on **Hugging Face Spaces**. A simple and intuitive **user interface (UI)** was developed, enabling business users to input parameters and obtain sales predictions instantly. This deployment ensures **network-wide accessibility**, empowering store managers and decision-makers with real-time forecasting capabilities.

Recommendation:

The **XGBoost** model was trained on a small dataset and deployed using a Python-based REST API, with a proof-of-concept (PoC) user interface hosted on Hugging Face Spaces. To productionize this setup, the model should be continuously retrained as new data becomes available, and the Python REST API should be integrated into a CI/CD pipeline to ensure seamless updates aligned with evolving business needs and enhanced its feature capability. The current solution serves PoC purposes only; for a production environment, additional features such as authentication, input validation, and integration with other business systems will be required to ensure robustness and scalability.

Business Problem Overview and Solution Approach

- SuperKart engaged a data science firm to develop a **sales forecast prediction model** using its historical sales data. The objective of the model is to help the business **optimize inventory management** and improve the **decision-making process** for building effective sales strategies. The company requires the model to be **easily accessible** to end users across its network of stores.
- To achieve this, the deployment must be **streamlined, automated, and robust**, ensuring it can handle **multiple users at scale** across different geographic locations. The solution should integrate seamlessly into SuperKart's **forecasting and decision-making processes**. Additionally, it must provide a **simple and intuitive user interface** for business users to input data and receive predictions. The final system should be **repeatable, scalable, and user-friendly**, enabling managers to leverage actionable insights efficiently. Overall, the challenge is to combine **advanced predictive analytics** with practical usability across the organization.

Business Problem Overview and Solution Approach

Following approach is used to build the solution

Data Analysis :

- Load historical sales data and validate the successful load by checking sample rows for correct columns and rows.
- Analyze statistical distribution of data, identify duplicates, and apply deduplication strategies.
- Detect missing values and apply corrective measures to address them.
- Perform **univariate analysis** using boxplots and histograms to study distributions, skewness, and outliers.
- Conduct **bivariate analysis** using correlation matrices to understand relationships between variables.
- Examine relationships of numeric and non-numeric columns with the target (prediction) column.

Data PreProcessing:

- Standardize categorical variables to reduce dimensionality.
- Analyze and simplify ID fields and other complex columns.
- Drop unnecessary or redundant columns.
- Split the dataset into training and testing sets.

Business Problem Overview and Solution Approach

Following approach is used to build the solution

Model Build and Evaluation

- Apply **one-hot encoding** for categorical variables.
- Shortlist and train multiple models, then fine-tune them for performance optimization.
- Evaluate models using error metrics and select the **best-performing model**.
- Serialize the final model for deployment.

Backend

- Flask API exposes REST endpoints accepting JSON inputs. Packaged in Docker for scalability and consistency.

Frontend

- Streamlit UI provides simple input interface. Abstracts model complexity for business users. Build A docker file for scalability and consistency.

Business Problem Overview and Solution Approach

Deployment

- Docker ensures portability and reliable multi-user access.
- Deployed on Hugging Face Spaces for secure, network-wide availability.
- Enables real-time predictions and actionable insights for decision-making

EDA Results

Data Volume and Product Characteristics

- **Data Size:** The dataset contains **8,763 rows** and **12 columns**.
- **Product Volume & Categories:** SuperKart carries **16 product categories**.
 - **High Volume Categories:** **Fruits & Vegetables** and **Snack Foods** are the largest categories, each accounting for approximately **13–14%** of the total product count.
 - **Low Volume Categories:** **Seafood** has the lowest product count.
- **Sugar Content:** The majority of products have **low sugar content**, accounting for **55%** of the total product count.
 - **Low Sugar Dominance:** **Fruits & Vegetables** and **Snack Foods** are **exclusively** comprised of Low Sugar items, indicating a strong market presence for this type of product across most food categories.
 - **Revenue:** Low Sugar items contribute the **highest revenue**, while products labeled 'reg' (regular) contribute the least.
 - There is strong need to standardize the sugar content categories to accurately assess 'Regular' sales
- **Pricing (MRP):** The average **Product MRP (Maximum Retail Price)** across most categories is around **\$150**.
- **Product Weight:** Product weights average around **12.5**. Low Sugar items show a consistent weight average of 12.5 but have many **outliers** in their weight distribution.
- **Product Allocated Area:** This metric is **left-skewed** with a mean around **0.687**.

[Link to Appendix slide on data background check](#)

EDA Results

Store and Location Profile

- **Store Count Distribution:** The data is split across four stores, but the volume is highly uneven:
 - **Store OUT004** accounts for the largest share of the data at **53.4%**.
 - **Store OUT002** accounts for the smallest share at only **13.1%**.
- **Store Size:** **Medium-sized stores** represent the majority, accounting for **68%** of the store volume, suggesting this is SuperKart's primary store format. Small-sized stores account for only 13%.
- **City Tier:** The majority of stores are located in a **Tier 2 city (71.5%)**. Tier 1 and Tier 3 cities each represent a smaller share (13–15%).
- **Store Type:** **Supermarket Type 2** is the dominant format, representing **53%** of all stores.
-

[Link to Appendix slide on data background check](#)

EDA Results

Overall Sales Contribution:

- **Product Category: Fruits & Vegetables** and **Snack Foods** contribute the **most to total revenue**. **Seafood** and **Breakfast** items contribute miniscule revenue.
- **Store Type/Location: Medium stores, Tier 2 cities, and Supermarket Type 2** locations collectively have the **highest revenue contribution**, which aligns with their majority representation in the store portfolio.

Store-Specific Performance:

- **OUT003:** Identified as **highly successful** with the highest overall product prices.
- **OUT002:** Identified as **struggling significantly** with the lowest typical prices, but it features a large number of outliers—very high-priced products placed alongside its predominantly budget-friendly inventory.

Store Size vs. Sales: **Large and Medium stores** are generally the **most successful** with high total sales, while **Small stores** typically struggle to achieve high sales volume

[Link to Appendix slide on data background check](#)
[Link to Appendix slide on Univariate And bivariate Analysis](#)

EDA Results

Correlation and Relationships

- **Strong Linear Relationships:**
 - **Product MRP and Product Weight** are **highly correlated**.
 - **Product Weight and Product Store Sales Total** show a **linear relationship**.
 - **Product MRP and Product Store Total Sales** show a **linear relationship**.
- **Weak/Nonlinear Relationships:**
 - **Other Variables:** Most other variables show **very weak correlation**, indicating they are not highly covariant.
 - **Product Store Sales Total:** This variable exhibits a **non-linear relationship** with other factors, though a pattern emerges when the Product Allocated Area is above 20% and sales are either very high (above 6,000) or very low (below 1,000).

[Link to Appendix slide on data background check](#)
[Link to Appendix slide on Univariate And bivariate Analysis](#)

Data Preprocessing

- Standardized Product Sugar content labels and change 'reg' to 'Regular' for consistency and remove fragmentation
- Product ID column prefix are FD, NC and DR represent food categories
- FD represent Food , DR represent drinks and NC represent health and hygiene and other items
- Sort age computed considering current year=2025 and subtracting stores establishment year
- Product type grouped into perishable and non-perishable categories
- Product weight , Allocated area , Product MRP and store sales total have many outlier but seems reasonable so no action needed
- Verified data duplication and missing values and found that there is no duplicate or missing values in the data set

Data Background and Contents

- Statistical summary of the data
- The dataset has 8,763 records with 12 columns consist of 5 numerical and 7 non-numerical columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8763 entries, 0 to 8762
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Product_Id                           8763 non-null   object
1   Product_Weight                       8763 non-null   float64
2   Product_Sugar_Content                8763 non-null   object
3   Product_Allocated_Area               8763 non-null   float64
4   Product_Type                         8763 non-null   object
5   Product_MRP                         8763 non-null   float64
6   Store_Id                             8763 non-null   object
7   Store_Establishment_Year             8763 non-null   int64
8   Store_Size                           8763 non-null   object
9   Store_Location_City_Type             8763 non-null   object
10  Store_Type                           8763 non-null   object
11  Product_Store_Sales_Total            8763 non-null   float64
dtypes: float64(4), int64(1), object(7)
memory usage: 821.7+ KB
```

Data Background and Contents

- Statistical summary of the data
- The dataset has 8,763 records with a mix of numeric and categorical features describing products, stores, and their total sales. Most fields have complete data, with numeric features like Product_Weight, Product_MRP, and Product_Store_Sales_Total, and categorical ones such as Product_Type, Store_Size, and Store_Type. Product_Id and Store_Id are unique identifiers and should not be used as model features.

data.describe(include="all").T

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	8763	8763	FD306	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	8763.0	NaN	NaN	NaN	12.653792	2.21732	4.0	11.15	12.66	14.18	22.0
Product_Sugar_Content	8763	4	Low Sugar	4885	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	8763.0	NaN	NaN	NaN	0.068786	0.048204	0.004	0.031	0.056	0.096	0.298
Product_Type	8763	16	Fruits and Vegetables	1249	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	8763.0	NaN	NaN	NaN	147.032539	30.69411	31.0	126.16	146.74	167.585	266.0
Store_Id	8763	4	OUT004	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	8763.0	NaN	NaN	NaN	2002.032751	8.388381	1987.0	1998.0	2009.0	2009.0	2009.0
Store_Size	8763	3	Medium	6025	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	8763	3	Tier 2	6262	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	8763	4	Supermarket Type2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	8763.0	NaN	NaN	NaN	3464.00364	1065.630494	33.0	2761.715	3452.34	4145.165	8000.0

Data Background and Contents

- Sample Data Head rows shows Product id has pattern of prefix of 2 alphanumeric char and followed by numbers
- Store type and store size and sugar content are categorical variable with storeld follows pattern of OUT[0-9]+ number format

	0	1	2	3	4
Product_Id	FD6114	FD7839	FD5075	FD8233	NC1180
Product_Weight	12.66	16.54	14.28	12.1	9.57
Product_Sugar_Content	Low Sugar	Low Sugar	Regular	Low Sugar	No Sugar
Product_Allocated_Area	0.027	0.144	0.031	0.112	0.01
Product_Type	Frozen Foods	Dairy	Canned	Baking Goods	Health and Hygiene
Product_MRP	117.08	171.43	162.08	186.31	123.67
Store_Id	OUT004	OUT003	OUT001	OUT001	OUT002
Store_Establishment_Year	2009	1999	1987	1987	1998
Store_Size	Medium	Medium	High	High	Small
Store_Location_City_Type	Tier 2	Tier 1	Tier 2	Tier 2	Tier 3
Store_Type	Supermarket Type2	Departmental Store	Supermarket Type1	Supermarket Type1	Food Mart
Product_Store_Sales_Total	2842.4	4830.02	4130.16	4132.18	2279.36

Data Background and Contents

- Sample Data tail rows

	8758	8759	8760	8761	8762
Product_Id	NC7546	NC584	NC2471	NC7187	FD306
Product_Weight	14.8	14.06	13.48	13.89	14.73
Product_Sugar_Content	No Sugar	No Sugar	No Sugar	No Sugar	Low Sugar
Product_Allocated_Area	0.016	0.142	0.017	0.193	0.177
Product_Type	Health and Hygiene	Household	Health and Hygiene	Household	Snack Foods
Product_MRP	140.53	144.51	88.58	168.44	224.93
Store_Id	OUT004	OUT004	OUT001	OUT001	OUT002
Store_Establishment_Year	2009	2009	1987	1987	1998
Store_Size	Medium	Medium	High	High	Small
Store_Location_City_Type	Tier 2	Tier 2	Tier 2	Tier 2	Tier 3
Store_Type	Supermarket Type2	Supermarket Type2	Supermarket Type1	Supermarket Type1	Food Mart
Product_Store_Sales_Total	3806.53	5020.74	2443.42	4171.82	2186.08

Model Building

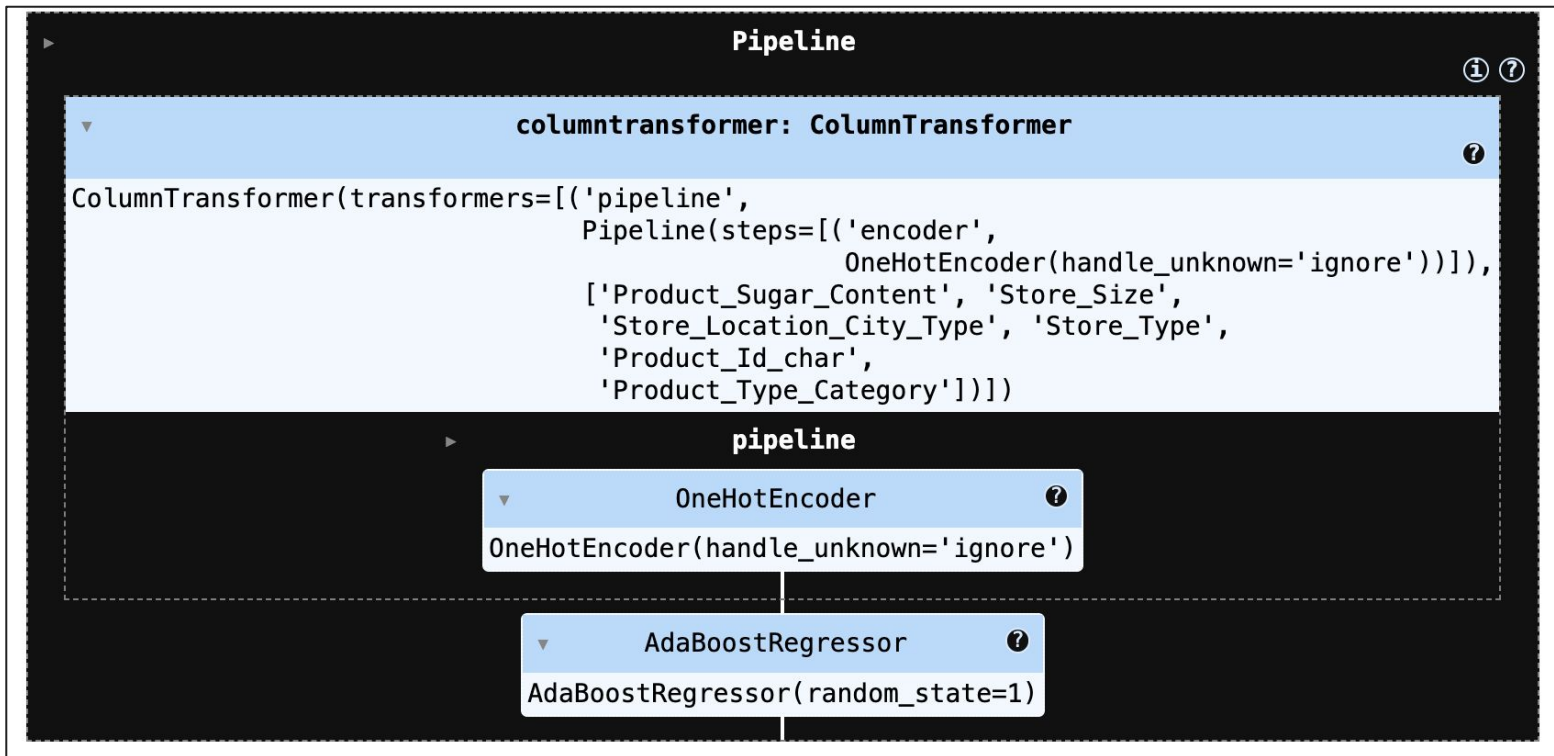
- Data preparation or Modeling
- Dropped the column Product_ID, {roduct_type , Store_Id and Store_Establishment_year (substituted by store Age)
- Derived feature column **Product_Type_Category with categorical representation of** perishable and nonperishable
 - Perishable items → ["Dairy","Meat", "Fruits and Vegetables","Breakfast","Breads","Seafood"]
 - Non Perishable item represent all other items
- Product Sales Total is prediction column so removed from the dependant feature and used as target variable for prediction
- Data is split in 70: 30 for Training vs Test dataset
- Training data got 6134 rows with 11 columns and test data has 2629 rows with 11 columns
- Apply One Hot encoding to categorical columns

Model Building

- Build function to compute the error value for model performance
- Following values are computed for the model performance RMSE , MAE , R squared Adj R squared and MAPE
- AdaBoost and XGBoost iteratively build sequential models that learn from and correct the errors (residuals) of the previous models, thereby reducing model **bias**, which often leads to higher accuracy than **bagging** methods (Random Forest and Bagging Regressor) and single models (Decision Tree)
- AdaBoost regression Model is Trained with hot encoding categorical column

Model Building

- AdaBoost Model Turned with preprocessing pipeline using OneHotEncoder and random state =1



Model Building

AdaBoost Training performance

AdaBoost Model Training					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	627.778405	512.191947	0.651694	0.651126	0.172981

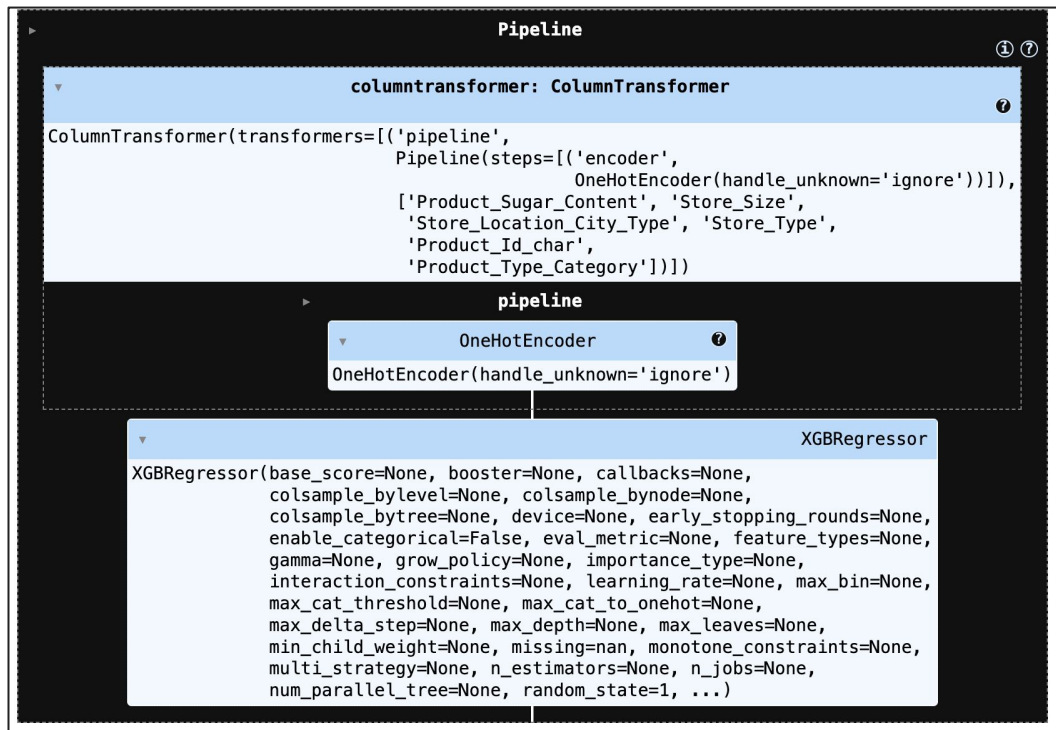
Adaboost Test performance

AdaBoost Model Test					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	647.483018	530.557049	0.633649	0.63225	0.193723

- **RMSE (627 on Train vs 647.48 Test)** – Indicates predictions deviate more from actual values.
- **MAE (512 on Train vs 530.56 Test)** – The average absolute error is quite large, confirming moderate accuracy.
- **R² (0.6510.511 on Train vs 0.6336 Test)** – Explains about **63% of the variance**, which is decent but not outstanding.
- **Adjusted R² (on Train vs 0.6322 Test)** – Similar to R², suggesting the model complexity is well-balanced.
- **MAPE (0.17 on Train vs 0.1937 Test)** – Roughly **19.4% average percentage error**, showing moderate performance.
- Model perform very similar on Test data set with slight degradation

Model Building

- XGBoost Model Turned with preprocessing pipeline using OneHotEncoder and random state=1



Model Building

- XGBoost Training performance

XGBoost Training					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	596.978222	468.965507	0.685033	0.684519	0.16569

- XGBoost Test performance

XGBoost Test					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	615.933034	485.429585	0.668482	0.667215	0.187421

XGBoost Model

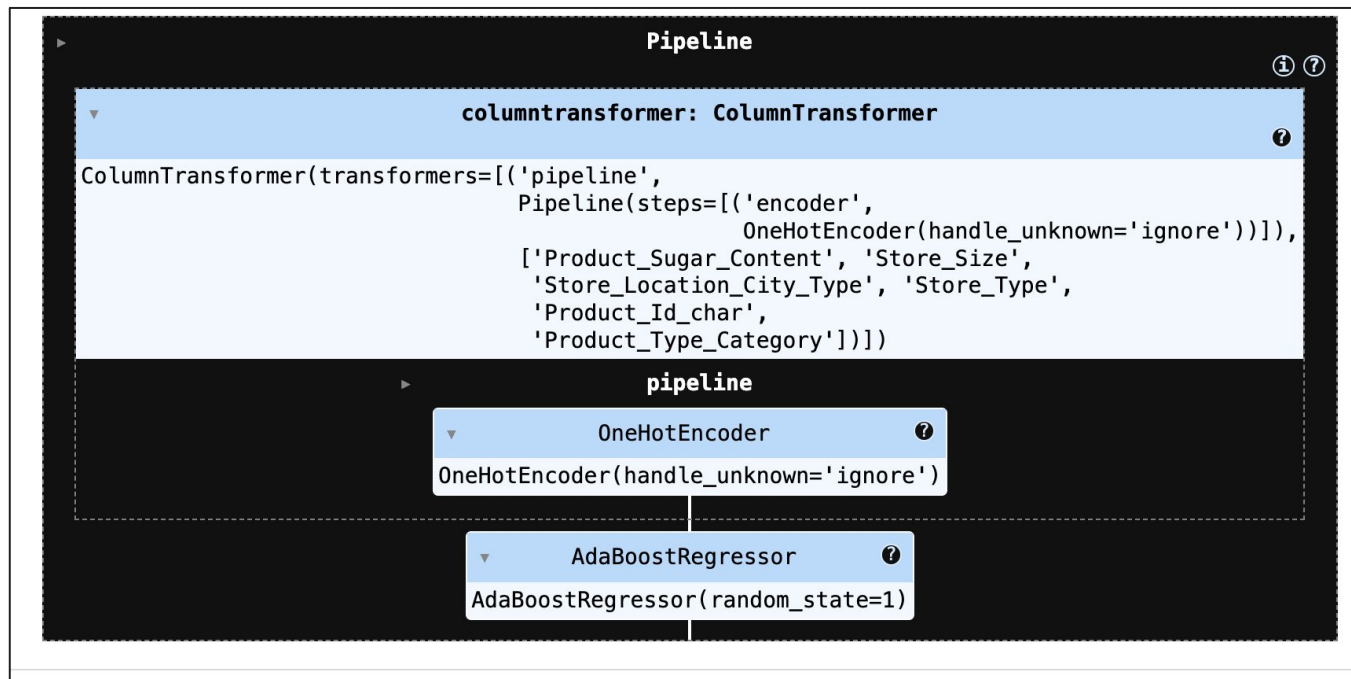
- **RMSE (596.97 on Train vs 615.93 Test)** – Significant improvement (~5% lower) over AdaBoost, meaning better predictive precision.
- **MAE(468.96 on Train vs 485.43 Test)** Errors are smaller on average; XGB captures patterns more effectively.
- **R² (0.685 on Train vs 0.668 Test)** variance indicate stronger model fit
- **Adjusted R² (0.684 on Train vs 0.667 Test)** is still high confirming efficiency despite model complexity
- **MAPE(0.16 on Train vs 0.18 Test)** improved to 18.7% accuracy and relatively better accuracy compare to AdBoost
- Model perform very similar on Test data set with slight degradation

Model Improvement - Hyperparameter Tuning

- Adaboost model tuned using gridSearchCV using below parameters

```
"adaboostregressor__n_estimators": [50, 100, 150, 200]
```

```
"adaboostregressor__learning_rate": [1.0, 0.1, 0.01, 0.001]
```



Model Building

- AdaBoost Tuned Training performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	618.174939	504.041133	0.662269	0.661663	0.171282

- Adaboost Tuned Test performance

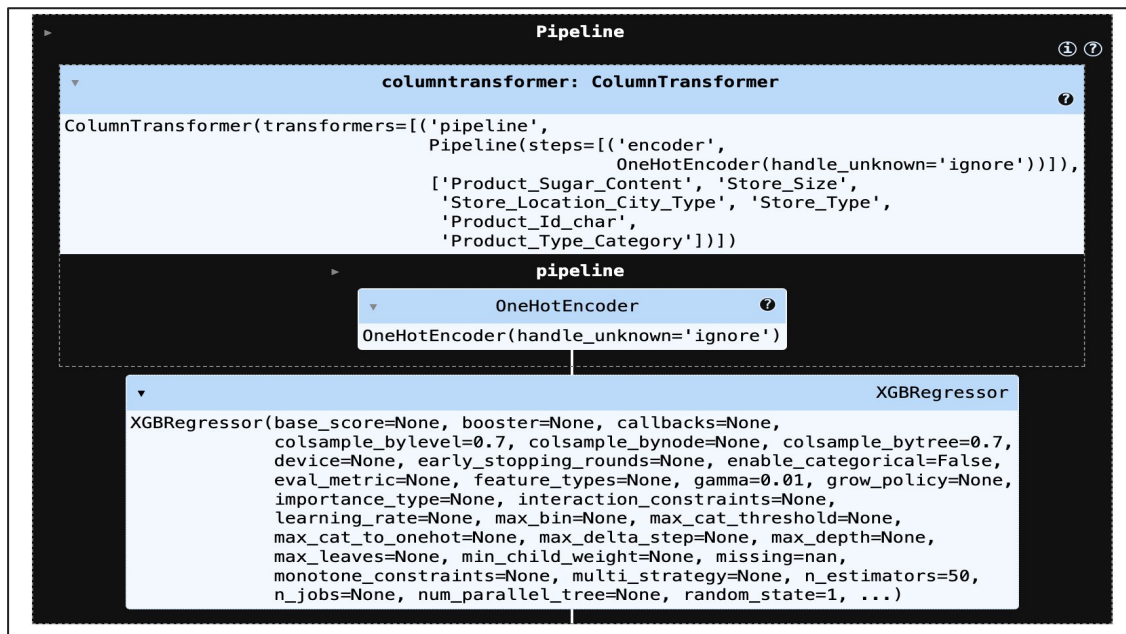
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	635.295258	520.981633	0.647311	0.645829	0.192216

All metrics are identical to baseline AdaBoost, implying tuning **did not yield any measurable gain** — possibly due to limited hyperparameter sensitivity or suboptimal tuning space.

Model Improvement - Hyperparameter Tuning

- XGboost model tuned using gridSearchCV using below parameters

```
"xgbregressor__n_estimators": [50, 100, 150, 200] "xgbregressor__subsample": [0.7, 0.8, 0.9, 1]
"xgbregressor__gamma": [0.01, 0.1, 1, 2] "xgbregressor__colsample_bytree": [0.7, 0.8, 0.9, 1.0]
"xgbregressor__colsample_bylevel": [0.7, 0.8, 0.9, 1.0]
```



Model Building

- XGBoost Tuned Training performance

XGBoost Tuned Train						
	RMSE	MAE	R-squared	Adj. R-squared	MAPE	
0	597.157033	468.551126	0.684845	0.68433	0.165697	

XGBoost Tuned Test performance

XGBoost Tuned Test						
	RMSE	MAE	R-squared	Adj. R-squared	MAPE	
0	615.737426	484.968315	0.668692	0.667427	0.187437	

RMSE (615.93) – Significant improvement (~5% lower) over AdaBoost, meaning better predictive precision.

MAE (484.96) – Errors are smaller on average; XGB captures patterns more effectively.

R² (0.669) – Explains nearly **67% of variance**, indicating stronger model fit.

Adjusted R² (0.6674) – Still high, confirming efficiency despite model complexity.

MAPE (0.1874) – Improved to **18.7%**, showing better relative accuracy.

Model Performance Summary

Training Performance Comparison

- XGBoost models lead: lowest RMSE (≈ 597), lowest MAE (≈ 468), highest R-squared and Adj R squared (≈ 0.68), and lowest MAPE (≈ 0.165), showing better fit and accuracy than AdaBoost throughout. AdaBoost models match each other and show higher RMSE/MAE and lower R-squared, confirming they're less accurate on the training data too.

Training performance comparison:

	ada_boost	ada_boost_tunes	xgb	xgb_tuens
RMSE	627.778405	618.174939	596.978222	597.157033
MAE	512.191947	504.041133	468.965507	468.551126
R-squared	0.651694	0.662269	0.685033	0.684845
Adj. R-squared	0.651126	0.661663	0.684519	0.684330
MAPE	0.172981	0.171282	0.165690	0.165697

Model Performance Summary

Test Performance Comparison

- XGBoost (xgb, xgb_tuens) achieves markedly lower RMSE (≈ 616), lower MAE (≈ 485), higher R-squared (≈ 0.668), and lower MAPE (≈ 0.187) compared to AdaBoost models (all metrics higher and R-squared ≈ 0.645).
- AdaBoost models (ada_boost, ada_boost_tunes) yield identical results, indicating no improvement from tuning, and consistently trail behind XGBoost on all metrics.
- XGBoost Tuend Model with parameters below parameter is selected as finalized at shown minimal error in all 4 model in test and train data set. XGBoost tuned slightly outperforms vanilla XGBoost, especially on test data (lowest RMSE, highest R-squared, lowest MAPE).

test performance comparison:

	ada_boost	ada_boost_tunes	xgb	xgb_tuens
RMSE	647.483018	635.295258	615.933034	615.737426
MAE	530.557049	520.981633	485.429585	484.968315
R-squared	0.633649	0.647311	0.668482	0.668692
Adj. R-squared	0.632250	0.645829	0.667215	0.667427
MAPE	0.193723	0.192216	0.187421	0.187437

Final Model

The best model is XGBoost tuned (xgb_tuens), showing the lowest test and training errors and highest R-squared values, indicating robust predictive power and generalization to unseen data.

- `XGBRegressor(base_score=None, booster=None, callbacks=None, colsample_bylevel=0.7, colsample_bynode=None, colsample_bytree=0.7, device=None, early_stopping_rounds=None, enable_categorical=False, eval_metric=None, feature_types=None, gamma=0.01, grow_policy=None, importance_type=None, interaction_constraints=None, learning_rate=None, max_bin=None, max_cat_threshold=None, max_cat_to_onehot=None, max_delta_step=None, max_depth=None, max_leaves=None, min_child_weight=None, missing=nan, monotone_constraints=None, multi_strategy=None, n_estimators=50, n_jobs=None, num_parallel_tree=None, random_state=1, ...)`

Deployment

- Backend Deployment details :

```
RepoUrl('https://huggingface.co/spaces/pshimpi/supercart', endpoint='https://huggingface.co',
repo_type='space', repo_id='pshimpi/supercart'), pr_revision=None, pr_num=None)
commit_url='https://huggingface.co/spaces/pshimpi/supercart/commit/e14f6576ebf8c3f3ab7bb76f997aa57a8a1438dd',
folder_path="backend_files"
```

Processing Files (1 / 1) : 100%  110kB / 110kB, 183kB/s

New Data Upload : 100%  110kB / 110kB, 183kB/s

...tent/backend_files/xgb_tuned.joblib: 100%  110kB / 110kB

```
CommitInfo(commit_url='https://huggingface.co/spaces/pshimpi/supercart/commit/e14f6576ebf8c3f3ab7bb76f997aa57a8a1438dd', commit_message='Upload folder
using huggingface_hub', commit_description='', oid='e14f6576ebf8c3f3ab7bb76f997aa57a8a1438dd', pr_url=None,
repo_url=RepoUrl('https://huggingface.co/spaces/pshimpi/supercart', endpoint='https://huggingface.co', repo_type='space', repo_id='pshimpi/supercart'),
pr_revision=None, pr_num=None)
```

Deployment

- Frontend Deployment details :

- `repo_url=RepoUrl('https://huggingface.co/spaces/pshimpi/supercartStreamlit',
endpoint='https://huggingface.co', repo_type='space', repo_id='pshimpi/supercartStreamlit'),
pr_revision=None, pr_num=None)`
- `commit_url='https://huggingface.co/spaces/pshimpi/supercartStreamlit/commit/06c9a7231866b9283b5f0af95e6f870b5df1d231'`
- `folder_path="frontend_files"`

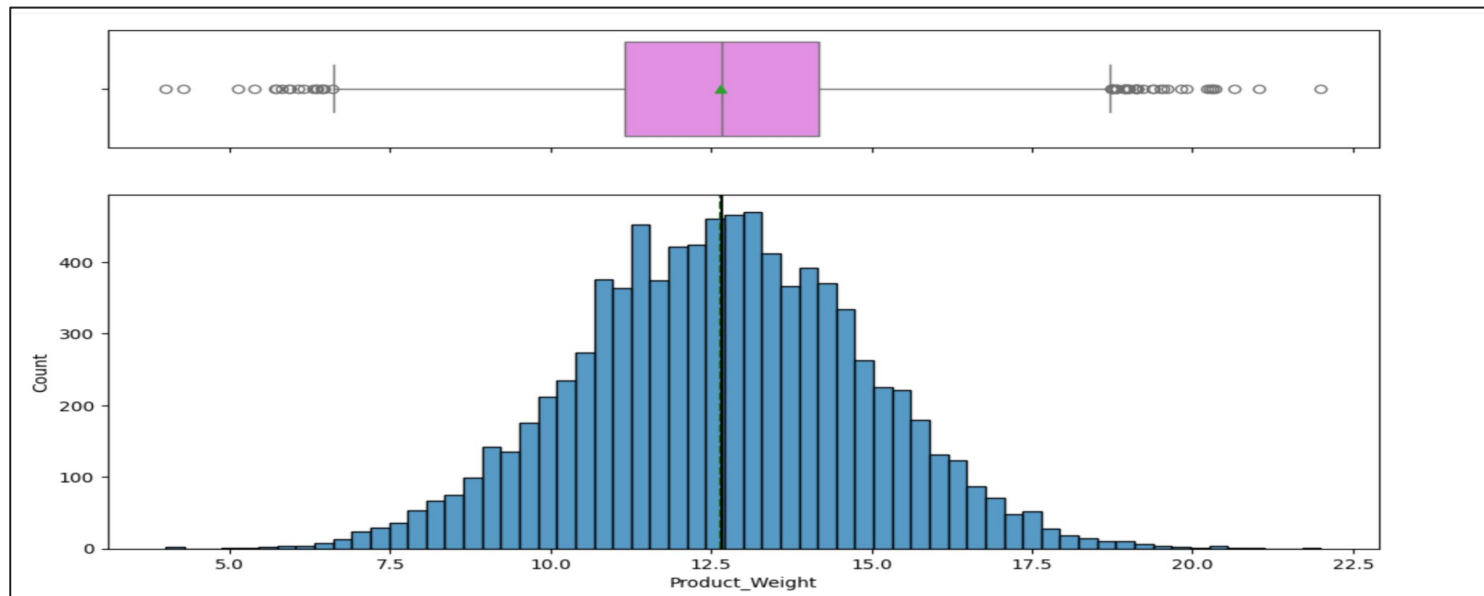
```
CommitInfo(commit_url='https://huggingface.co/spaces/pshimpi/supercartStreamlit/commit/06c9a7231866b9283b5f0af95e6f870b5df1d231',  
commit_message='Upload folder using huggingface_hub', commit_description='', oid='06c9a7231866b9283b5f0af95e6f870b5df1d231', pr_url=None,  
repo_url=RepoUrl('https://huggingface.co/spaces/pshimpi/supercartStreamlit', endpoint='https://huggingface.co', repo_type='space',  
repo_id='pshimpi/supercartStreamlit'), pr_revision=None, pr_num=None)
```


Appendix

Additional slides on Data Analysis to show Univariate Analysis and Bivariate Analysis added

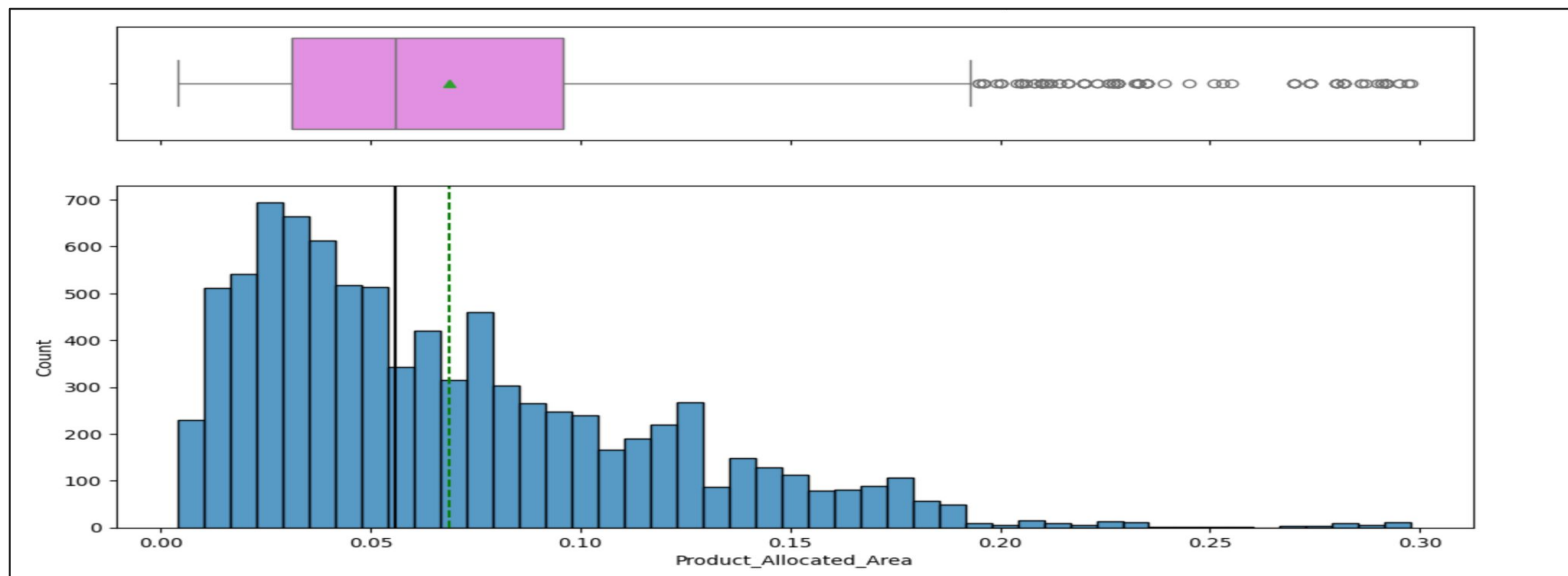
Univariate Analysis

- Product_Weight: is balanced data with many outlier but seems legit data from business perspective
- With mean = 12.65 , min=4.0 and max = 22.0



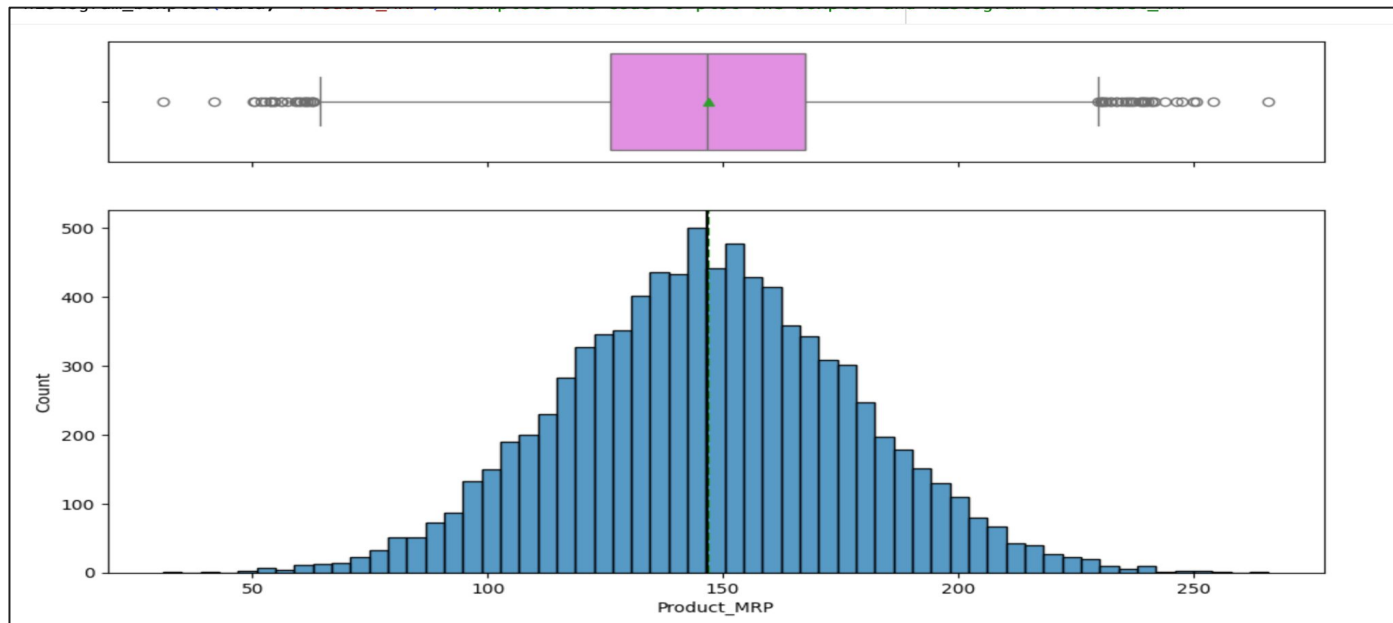
Univariate Analysis

- Product_allocated_Area: data is left skewed with many outlier on the higher end perspective
- With mean = 12.0.0687 , min=0.004 and max = 0.298



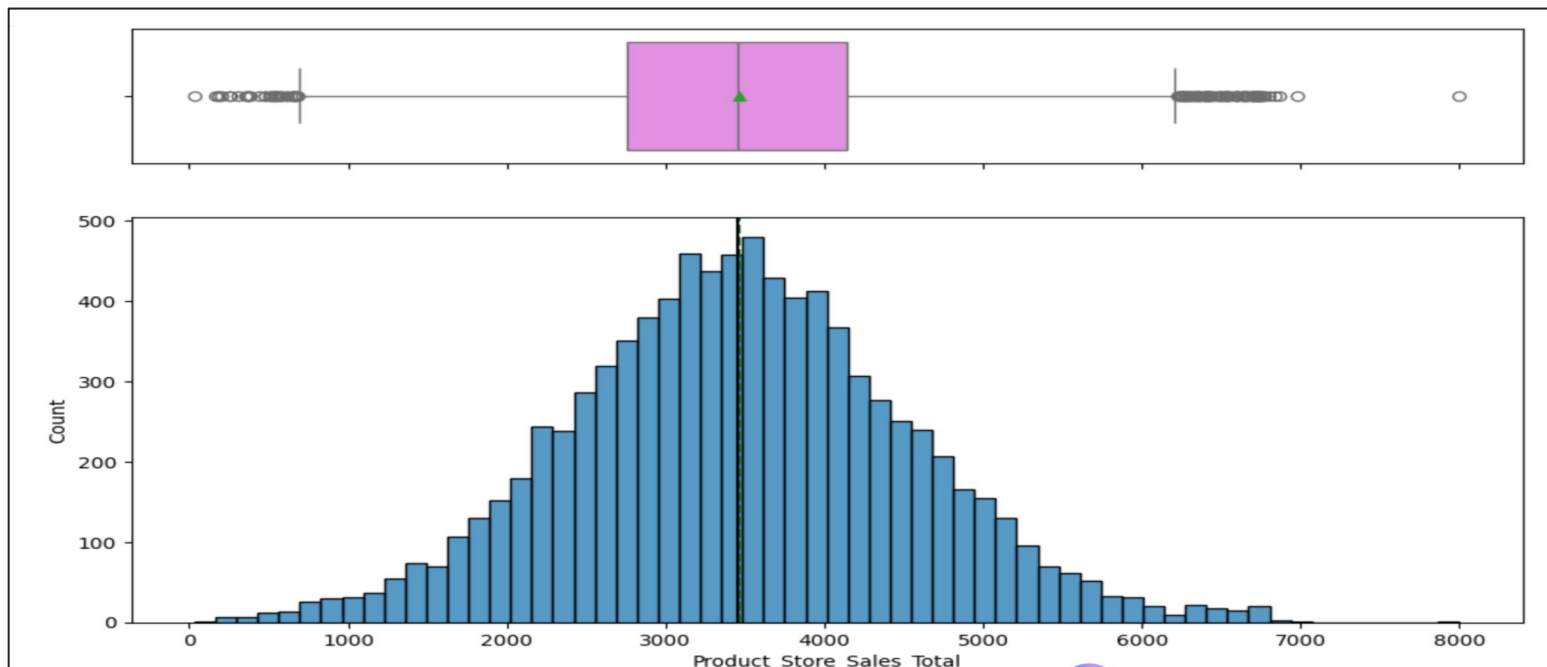
Univariate Analysis

- Product_MRP: is balanced data with many outlier but seems legit data from business perspective
- With mean = 147.0 , min=31.0 and max = 266.0 with 75% product MRP below 167.58



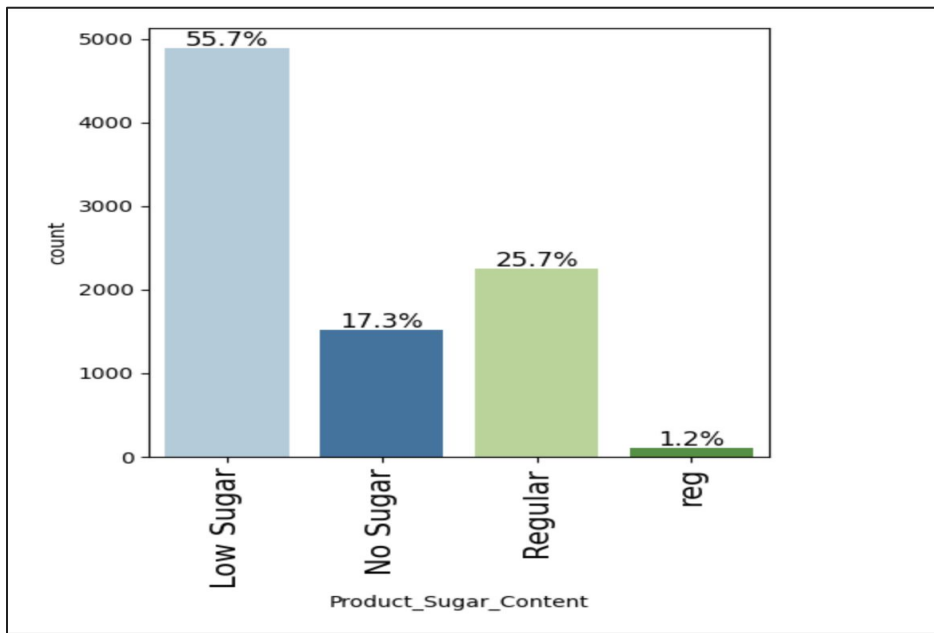
Univariate Analysis

- Product_Store_Sales_Total: is balanced data with many outlier but seems legit data from business perspective
- With mean = 3464 , min=33.0 and max = 8000



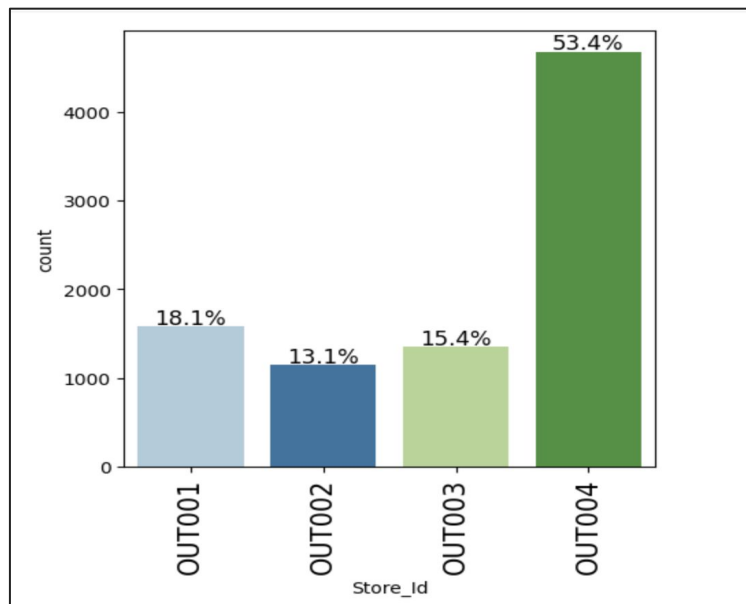
Univariate Analysis

- Product_Sugar_Content: Data indicate there is need for standardizing the categories to avoid unnecessary fragmentation. Also Low sugar product account for more than half the product and no sugar is account only for 17% and rest product are has regular sugar content



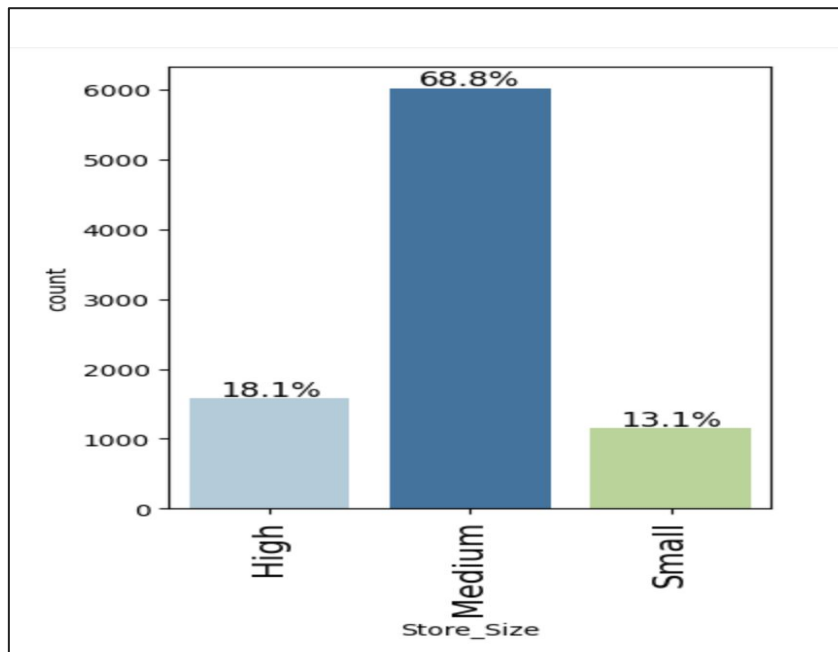
Univariate Analysis

- Store_ID: There are four stores with id from OUT-0001 to OUT -0004
- Store ID OUT004 account for 53% share and other stores has 13-18% share



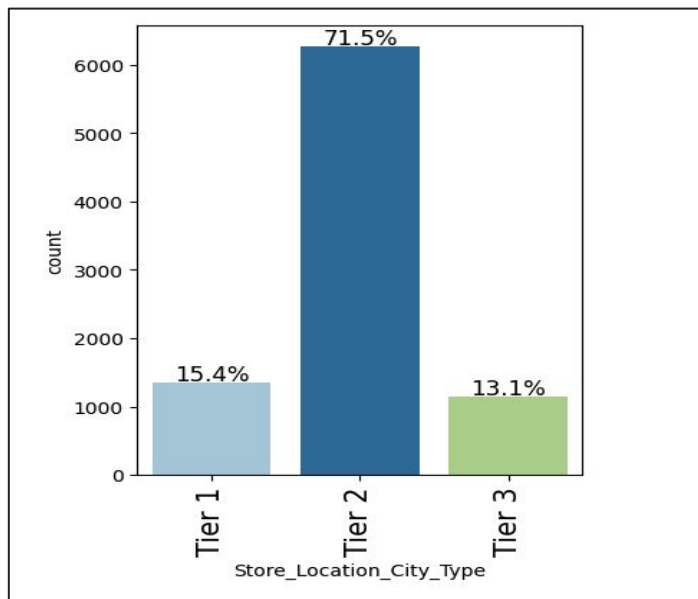
Univariate Analysis

- Store_size: data is split into 3 categories High , Medium and small ,
- Small stores has 68.8 % shar and followed by High at 18.1 and small at 13.1, Indicating Superkart has product concentrated in Medium size stores



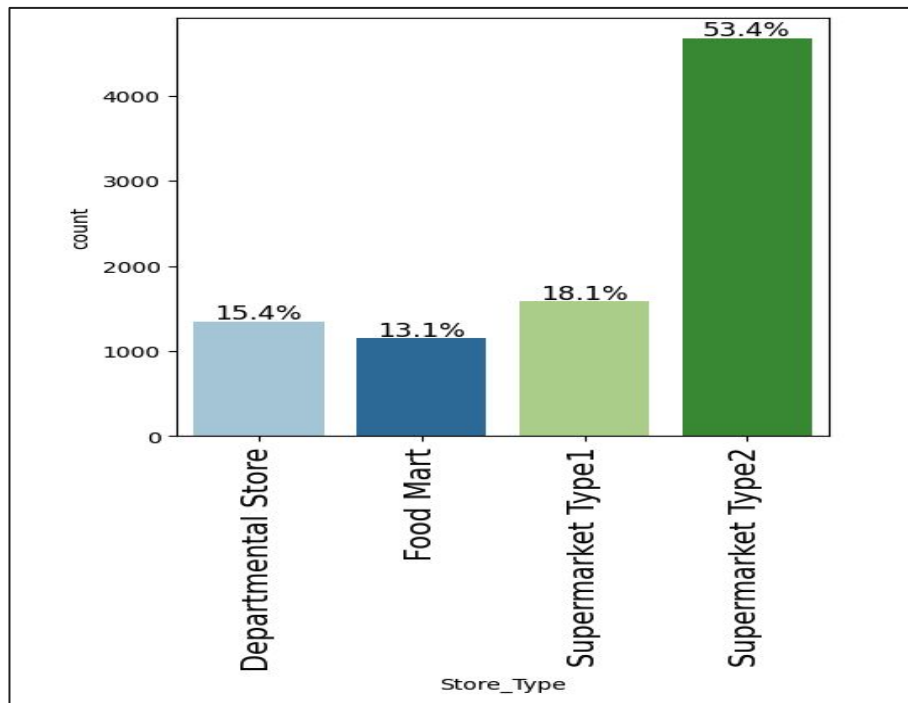
Univariate Analysis

- `Store_Location_City_Type`: Data has 3 categories Tier 1, 2 and 3. Tier 2 has majority share at 71.5% percent and rest are in Tier 1 and Tier 3.



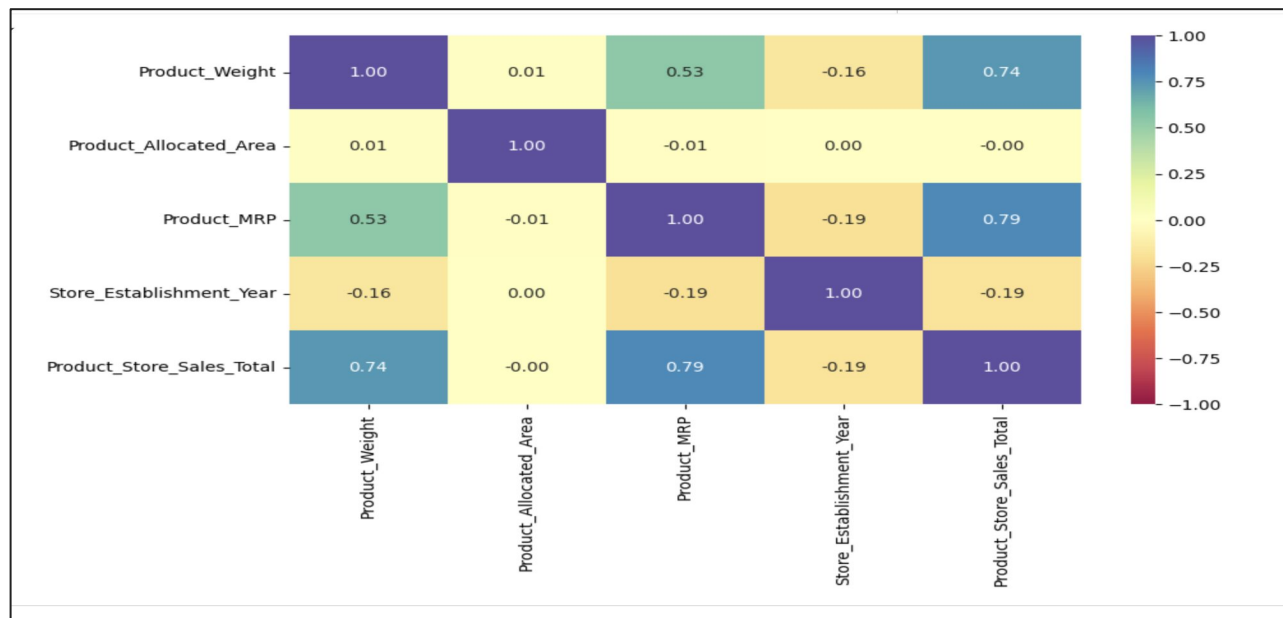
Univariate Analysis

- Store Type: SuperKart have multiple store types Department stores , food mart , Super market type 1 and Supermarket type 2
- Supermarket type 2 store account for more than half the share and other types similar share ranging from 13 to 18 %



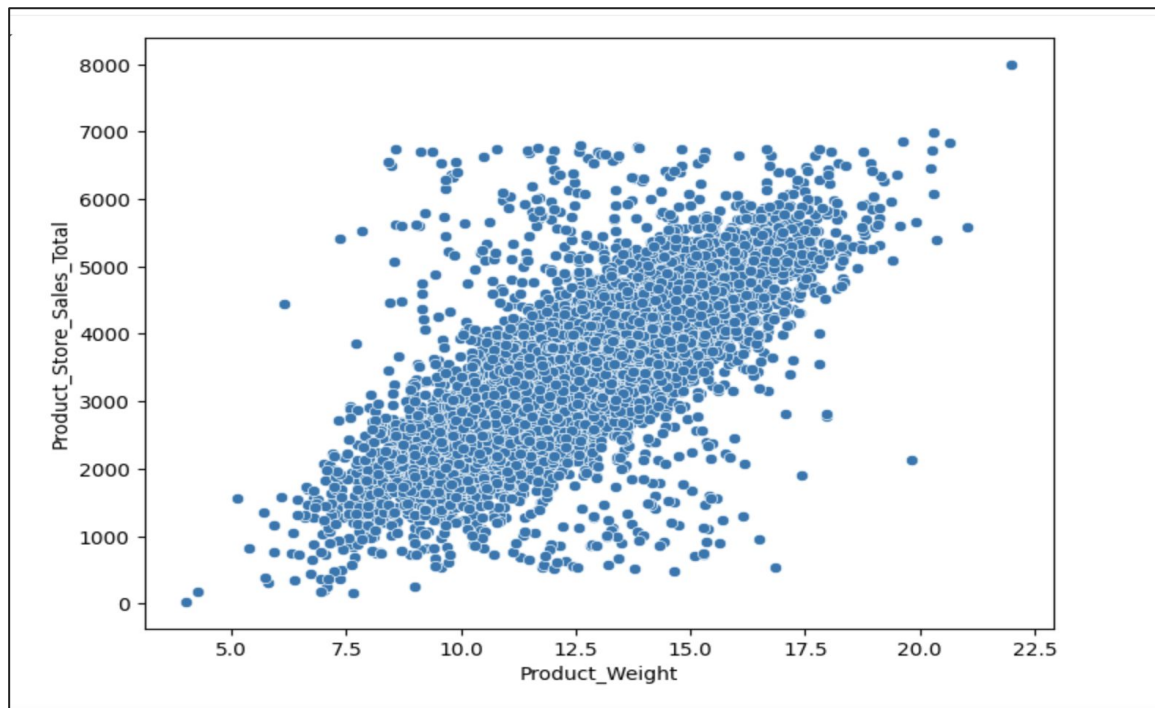
Bivariate Analysis

- Correlation Matrix shows strong +ve correlations between Product_Store_Sales_Total and Product_MRP, Product Weight and Product_Store_Sales_Total also has strong +ve correlations
Other variable seems independent feature and good candidate for model training



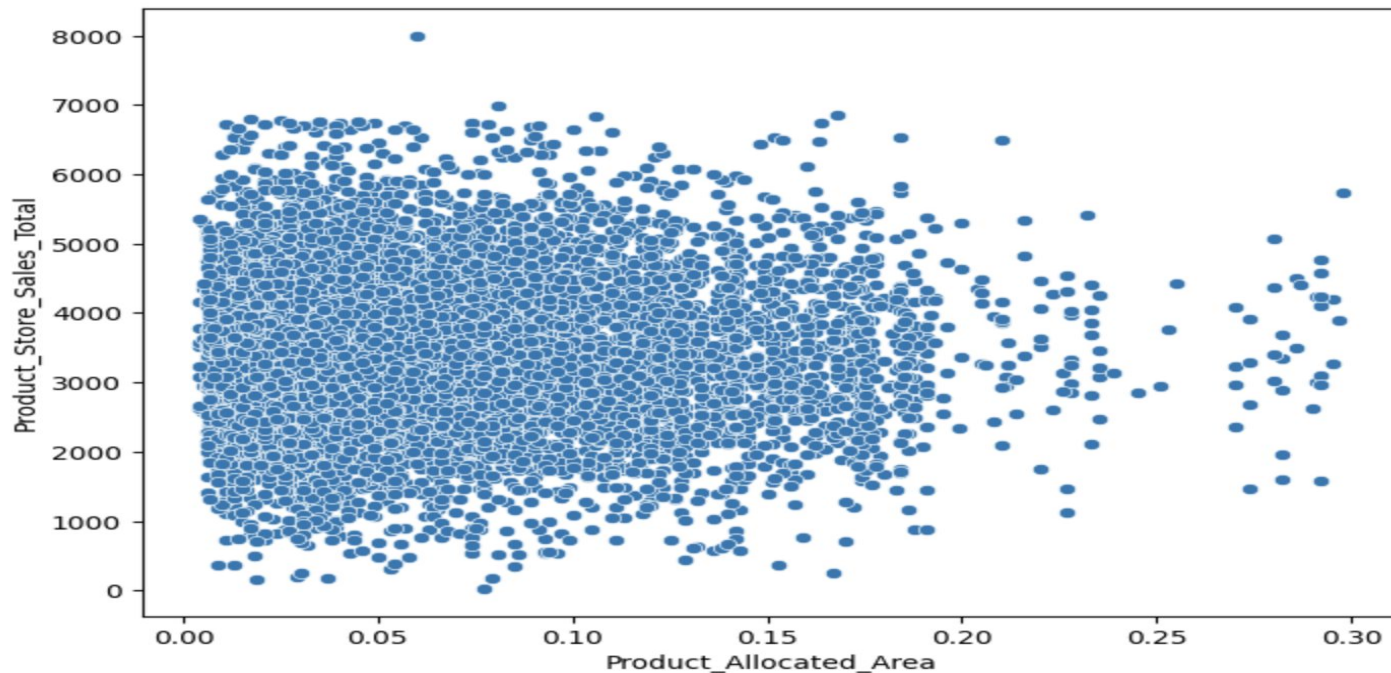
Bivariate Analysis

- Product weight and Product store sales total have linear relationship can be seen in below scatterplot



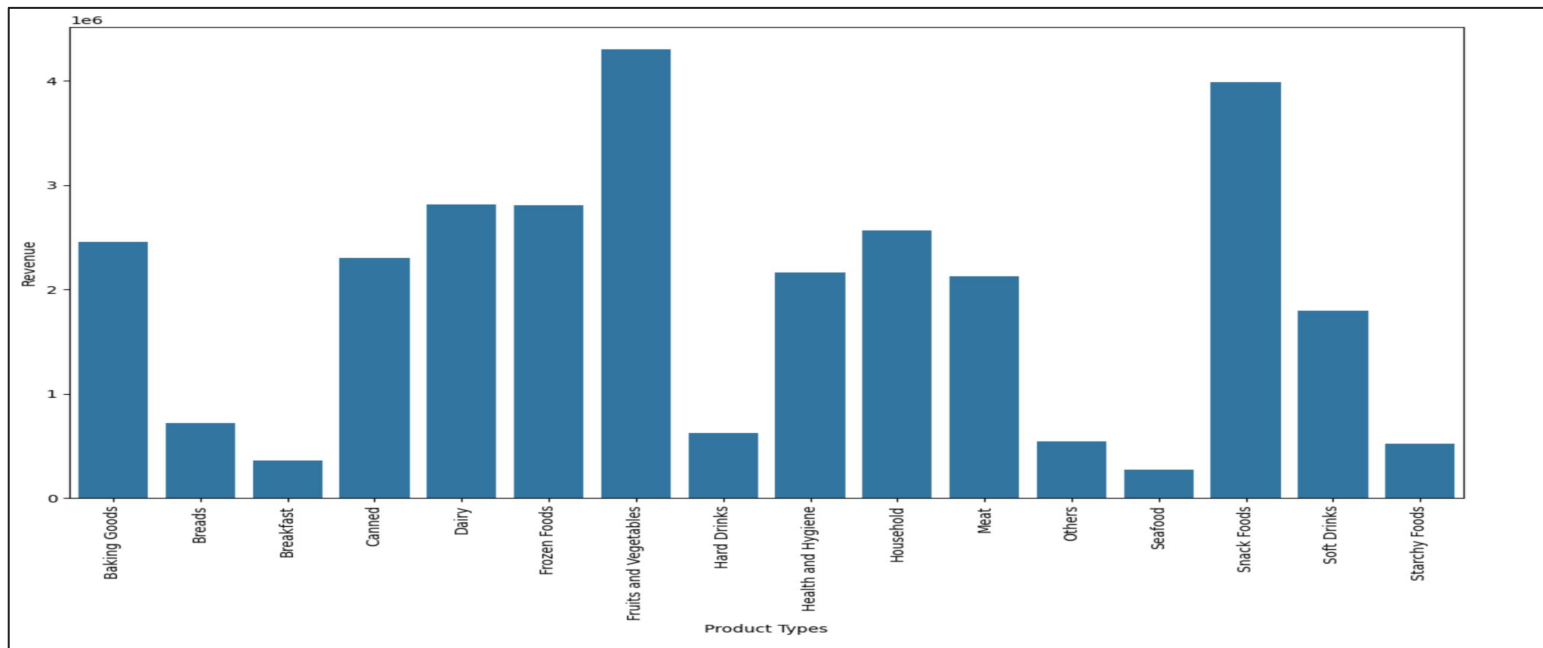
Bivariate Analysis

- Product Sales total and Product Allocated has weak correlation and don't show direct relationship but major sales volume concentrated for product area below 0.20



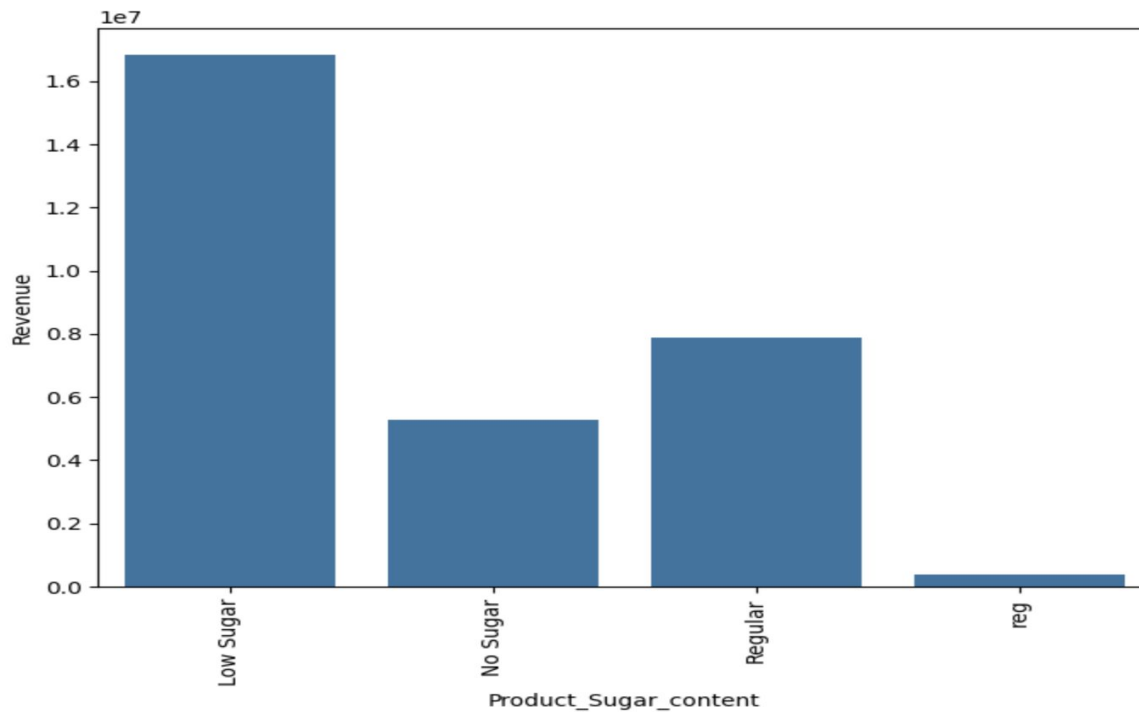
Bivariate Analysis

- Product Type and Revenue shows that snack foods and Fruits & vegetable account maximum revenue around 4.5×10^6 seafood, starchy foods, breads and breakfast items contribute very less in revenue/Sales Total around 0.5×10^6 to 0.8×10^6



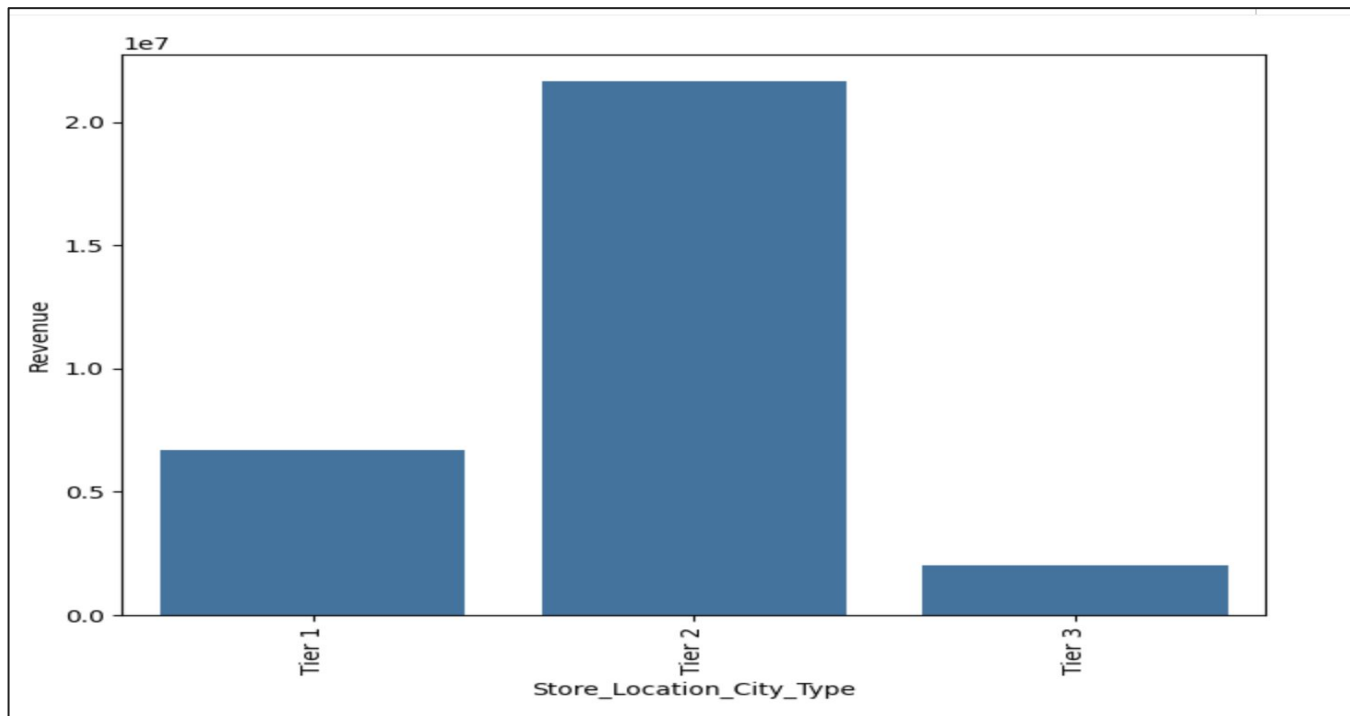
Bivariate Analysis

- Sugar Content and Revenue shows that low Sugar content item contribute around 1.6×10^7 and No sugar item around 0.55×10^7 in revenue. Also Regular and reg seems same categories and should be merged



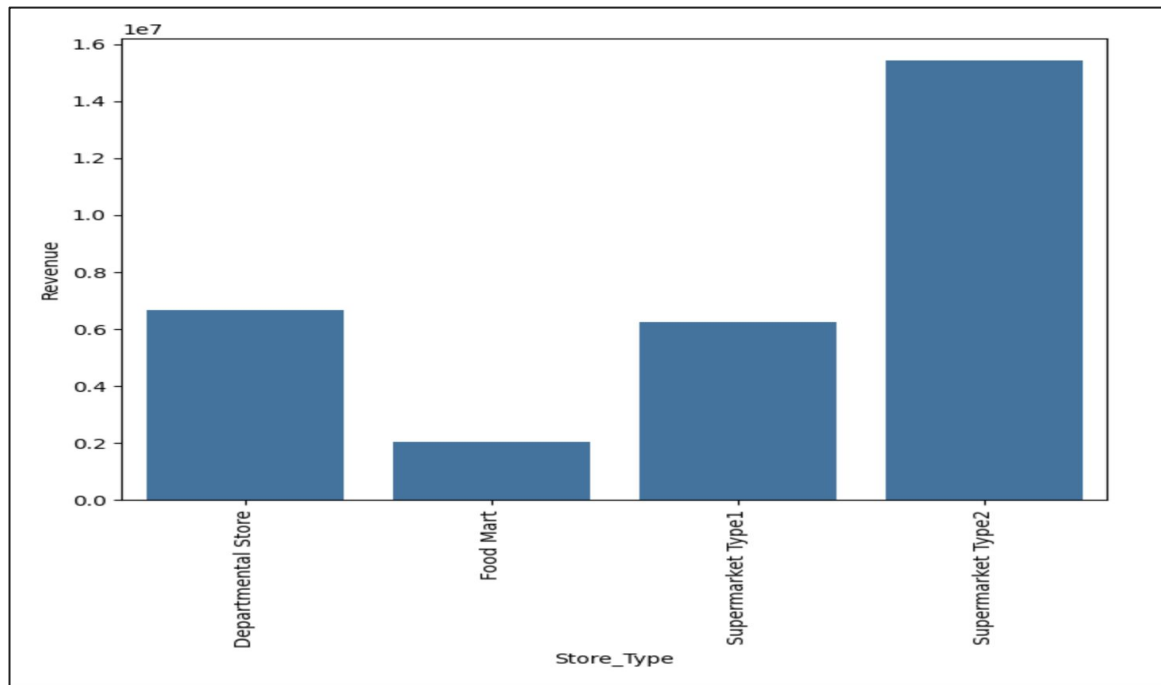
Bivariate Analysis

- Store Location City type shows that Tier 2 location account for 2.4×10^7 revenue and Tier 1 account for 0.6×10^7 and 0.3×10^7 revenue



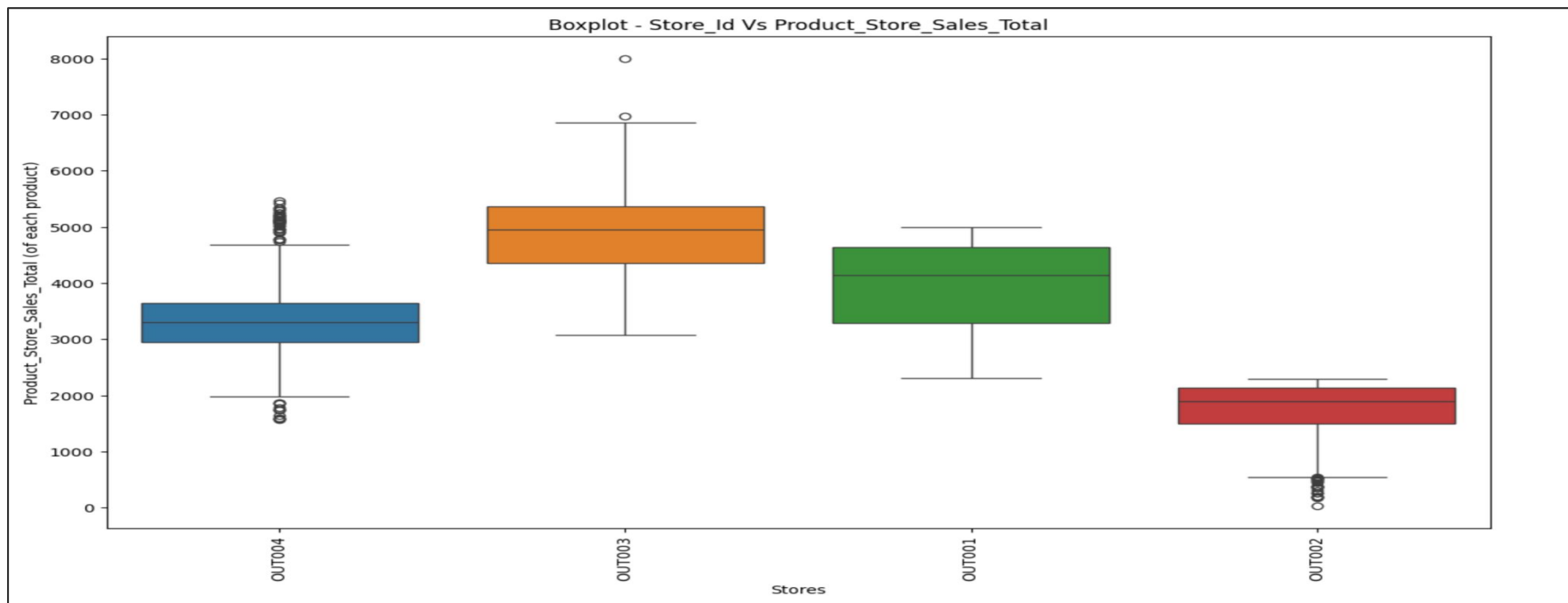
Bivariate Analysis

- Supermarket Type 2 account for 1.58×10^7 revenue and Department store and Supermarket has around 0.6×10^7 revenue. Food market account for only and 0.2×10^7 revenue.



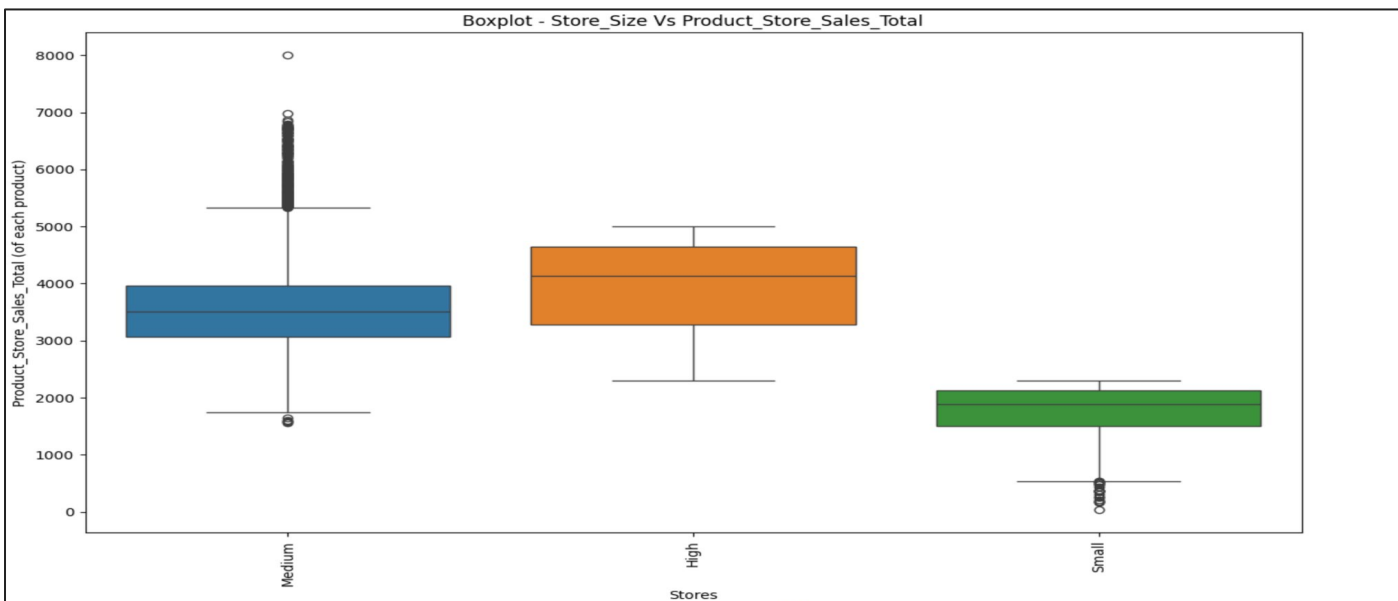
Bivariate Analysis

Store vs Sales bar plot indicate: store OUT003 This store has the highest median product sales, at approximately 4,900. Highest core performance; has the single best-selling product. OUT001 had Strong performance with a stable, wide sales range. Stoe OUT004 had Moderate, consistent sales, with several high-performing 'star' products. OUT004 Significantly lower sales across the board; many products perform poorly.



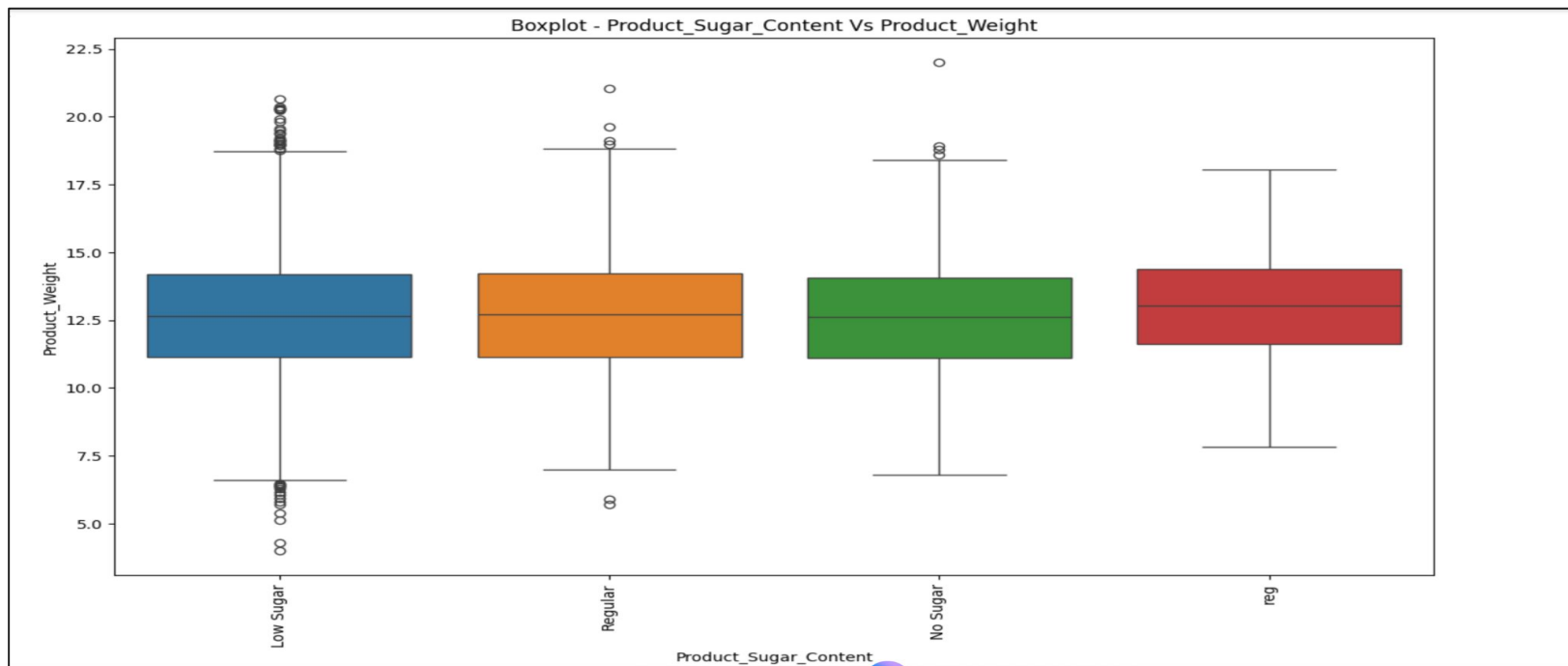
Bivariate Analysis

- Store Size vs Sales indicate High Store size performed better than other with mean product sales greeted that \$4000 which is significantly higher than others and small store struggle a lot and many product don't have much sales and there are no high performing product but Medium size store had quite significant product that perform quite well in terms of sales and max went till 8K



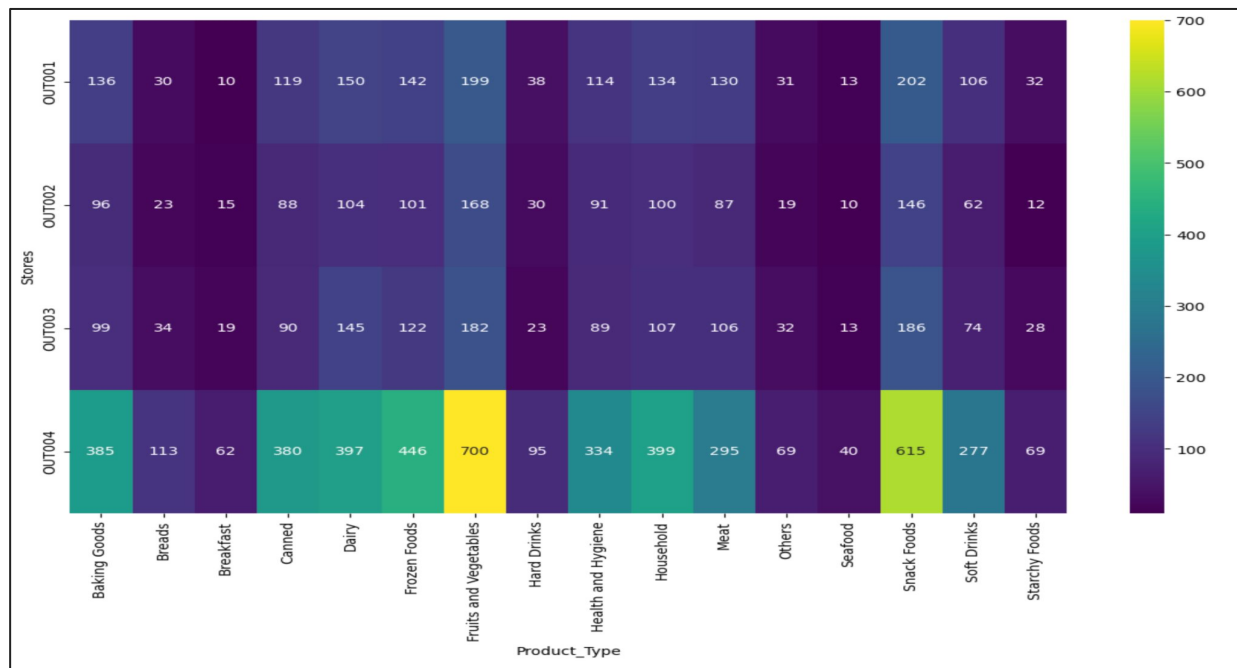
Bivariate Analysis

Product sugar content and weight has similar distribution avg around 12.5 and Low sugar content item has many outlier



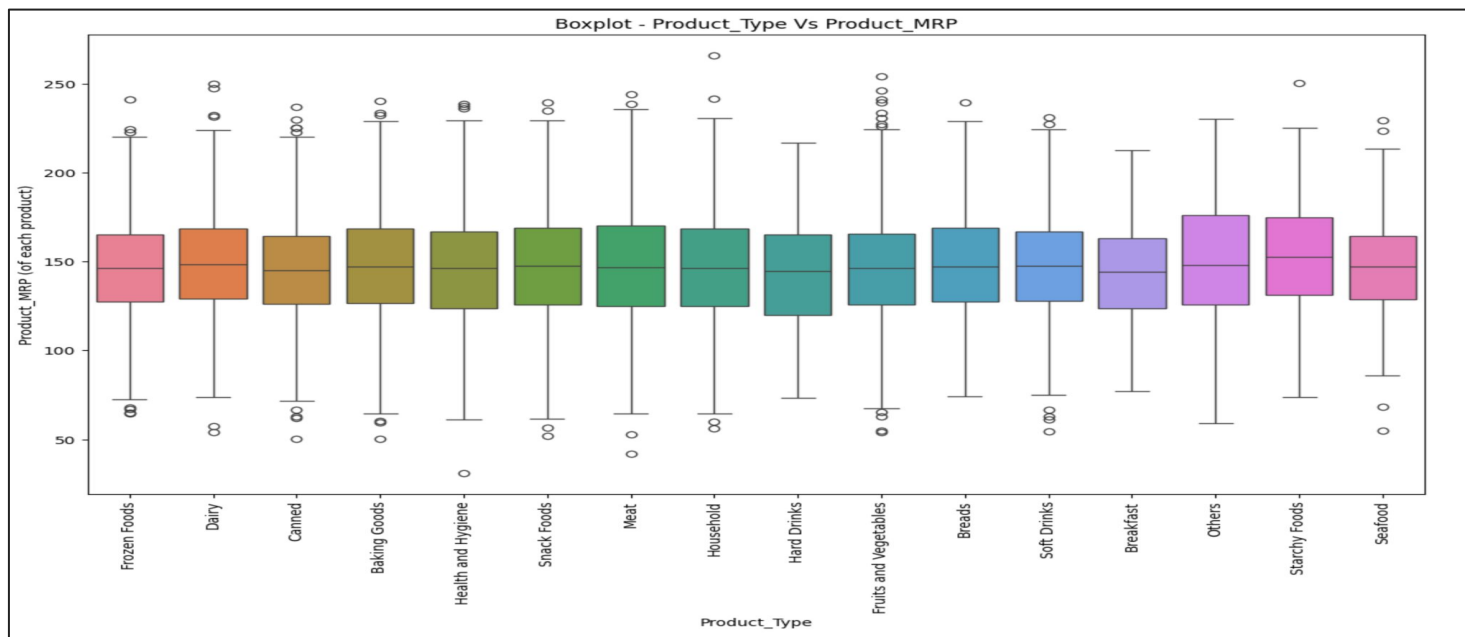
Univariate Analysis

- Product count by store shows that Fruits & vegetables has the largest products present (700) and followed by Snackfoods (615). OUT0004 store has comparatively larger number of product Type compare to others. Store OUT0003 and OUT0002 have similar product type count and store OUT001 has slightly largest product types available in each category



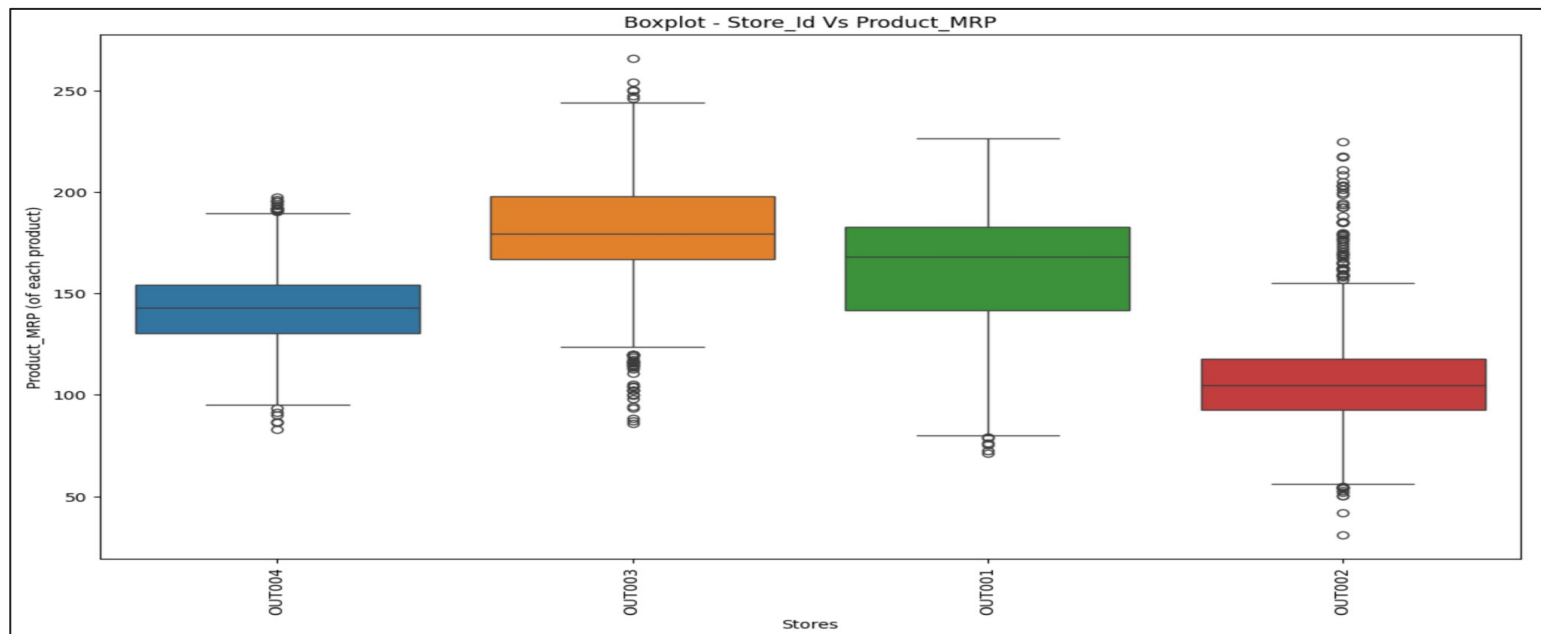
Univariate Analysis

- Product Type and Product MRP box plot indicate Product MRP mean prices are similar around `50 Hard Drinks have many outlier and max goes upto 250 similarly health and Hygiene has some product price goes around 10 which is least among all the product type



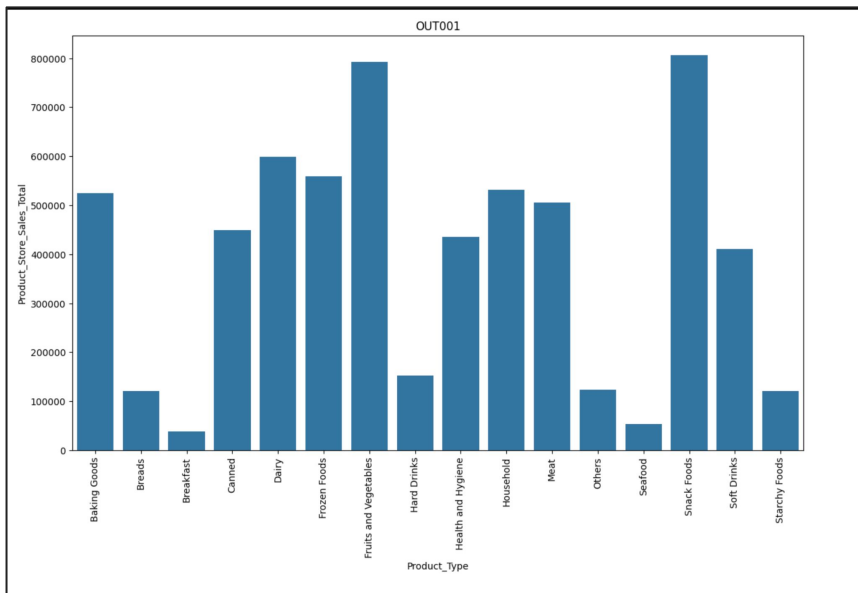
Bivariate Analysis

- Store OUT003 seems to have premium product indicated by higher mean compare to others This store cater to wide variety of product price range and had some outlier in higher price point and also in lower price point. Sore OUT002 has the lowest mean indicating it has most of the product at lower price point but also has some high end product as well. Sroe OUT001 and is moderately below store OUT003 in pricer point and Sote OUT004 is slight have higher priced product than store OUT002



Store OUT0001

Store OUT0001 cater to wide variety of product categories and it highest sales is around 800K in Fruits & vegetable and Snack foods. Breakfast seafood sales is quite low and is below. It is located in tier 2 city

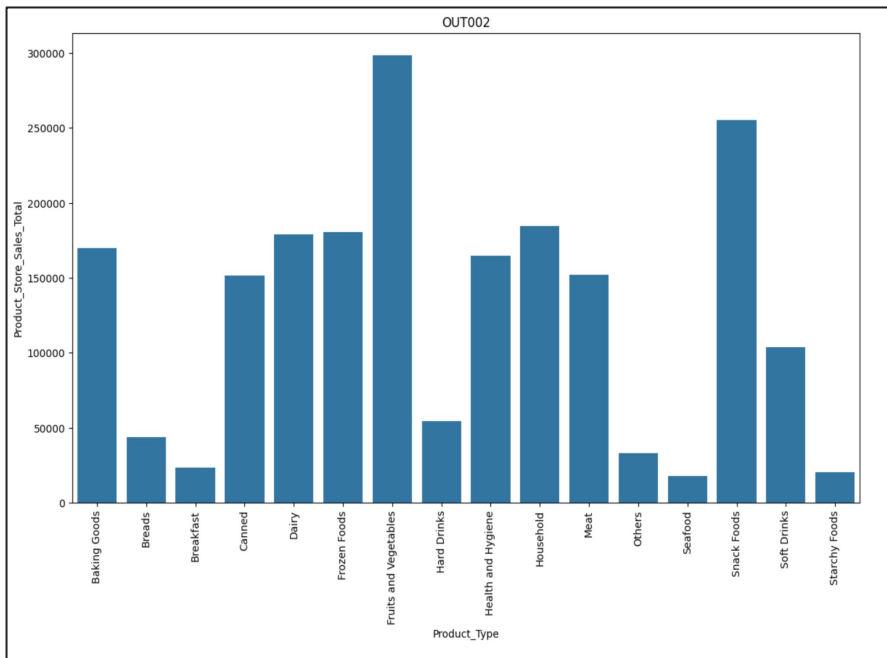


```
data.loc[data["Store_Id"] == "OUT001"].describe(include="all").T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1586	1586	NC7187	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1586.0	NaN	NaN	NaN	13.458865	2.064975	6.16	12.0525	13.96	14.95	17.97
Product_Sugar_Content	1586	4	Low Sugar	845	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1586.0	NaN	NaN	NaN	0.068768	0.047131	0.004	0.033	0.0565	0.094	0.295
Product_Type	1586	16	Snack Foods	202	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1586.0	NaN	NaN	NaN	160.514054	30.359059	71.35	141.72	168.32	182.9375	226.59
Store_Id	1586	1	OUT001	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1586.0	NaN	NaN	NaN	1987.0	0.0	1987.0	1987.0	1987.0	1987.0	1987.0
Store_Size	1586	1	High	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1586	1	Tier 2	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1586	1	Supermarket Type1	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	1586.0	NaN	NaN	NaN	3923.778802	904.62901	2300.56	3285.51	4139.645	4639.4	4997.63

Store OUT0002

Store OUT0002 cater to wide variety of product categories and it highest sales is around 300K in Fruits & vegetable and Snack foods account for 250K . Breakfast seafood sales is quite low and is below. It is located in tier 3 city and has lesser sales compare to store OUT0001



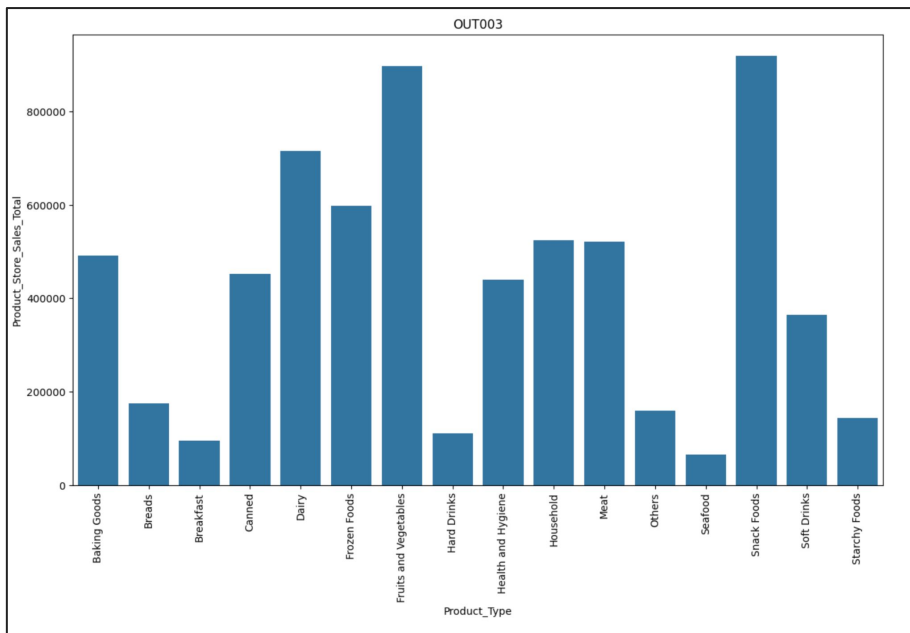
OUT002

```
data.loc[data["Store_Id"] == "OUT002"].describe(include="all").T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1152	1152	NC2769	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1152.0	NaN	NaN	NaN	9.911241	1.799846	4.0	8.7675	9.795	10.89	19.82
Product_Sugar_Content	1152	4	Low Sugar	658	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1152.0	NaN	NaN	NaN	0.067747	0.047567	0.006	0.031	0.0545	0.09525	0.292
Product_Type	1152	16	Fruits and Vegetables	168	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1152.0	NaN	NaN	NaN	107.080634	24.912333	31.0	92.8275	104.675	117.8175	224.93
Store_Id	1152	1	OUT002	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1152.0	NaN	NaN	NaN	1998.0	0.0	1998.0	1998.0	1998.0	1998.0	1998.0
Store_Size	1152	1	Small	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1152	1	Tier 3	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1152	1	Food Mart	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	1152.0	NaN	NaN	NaN	1762.942465	462.862431	33.0	1495.4725	1889.495	2133.6225	2299.63

Store OUT0003

- Store OUT0003 cater to wide variety of product categories and it highest sales is around 860K in Fruits & vegetable and Snack foods account for 880K . Breakfast seafood sales is quite low and is below 10K. It is located in tier 1 city and has better sales compare to store OUT001 and is best performing store



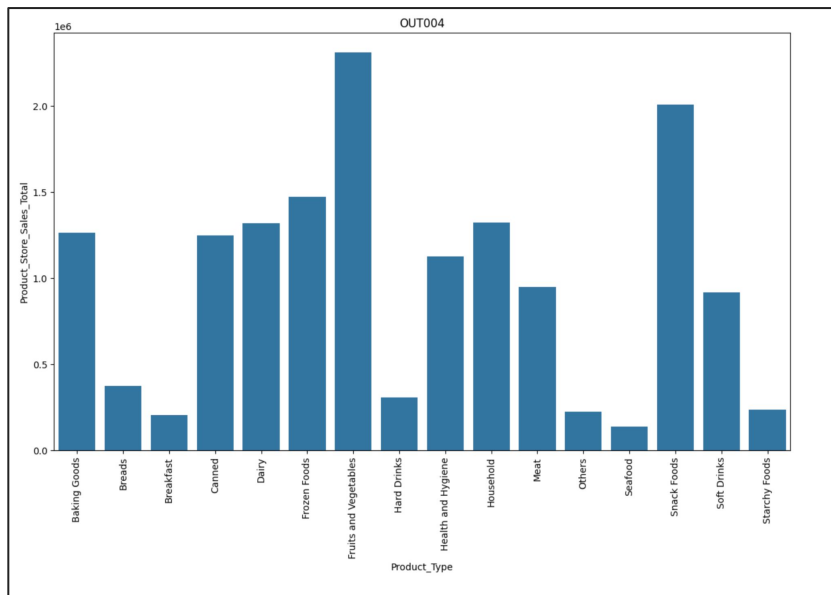
```
data.loc[data["Store_Id"] == "OUT003"].describe(include="all", T
```

(+)

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1349	1349	NC522	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1349.0	NaN	NaN	NaN	15.103692	1.893531	7.35	14.02	15.18	16.35	22.0
Product_Sugar_Content	1349	4	Low Sugar	750	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1349.0	NaN	NaN	NaN	0.068637	0.048708	0.004	0.031	0.057	0.094	0.298
Product_Type	1349	16	Snack Foods	186	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1349.0	NaN	NaN	NaN	181.358725	24.796429	85.88	166.92	179.67	198.07	266.0
Store_Id	1349	1	OUT003	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1349.0	NaN	NaN	NaN	1999.0	0.0	1999.0	1999.0	1999.0	1999.0	1999.0
Store_Size	1349	1	Medium	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1349	1	Tier 1	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1349	1	Departmental Store	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	1349.0	NaN	NaN	NaN	4946.966323	677.539953	3069.24	4355.39	4958.29	5366.59	8000.0

Univariate Analysis

- Store OUT0003 cater to wide variety of product categories and it highest sales is around 250K in Fruits & vegetable and Snack foods account for 200K . Breakfast seafood sales is quite low and is below 10K. It is located in tier 2 city and has better sales compare to store OUT001 and is best performing store



```
data.loc[data["Store_Id"] == "OUT004"].describe(include="all").T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	4676	4676	NC584	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	4676.0	NaN	NaN	NaN	12.349613	1.428199	7.34	11.37	12.37	13.3025	17.79
Product_Sugar_Content	4676	4	Low Sugar	2632	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	4676.0	NaN	NaN	NaN	0.069092	0.048584	0.004	0.031	0.056	0.097	0.297
Product_Type	4676	16	Fruits and Vegetables	700	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	4676.0	NaN	NaN	NaN	142.399709	17.513973	83.04	130.54	142.82	154.1925	197.66
Store_Id	4676	1	OUT004	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	4676.0	NaN	NaN	NaN	2009.0	0.0	2009.0	2009.0	2009.0	2009.0	2009.0
Store_Size	4676	1	Medium	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	4676	1	Tier 2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	4676	1	Supermarket Type2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	4676.0	NaN	NaN	NaN	3299.312111	468.271692	1561.06	2942.085	3304.18	3646.9075	5462.86



Happy Learning !

